

Problem 1 Lexing and Parsing

Assume we are working over alphabet $\Sigma = \{\text{death, thou, fair, noble, sword}\}$. Let L be the language consisting of sentences which obey the following conditions:

- Each sentence must begin with thou.
- The word death must always be immediately preceded by the word fair.
- Each sentence may contain the word noble at most once (0 or 1 times).

For example, “thou noble sword”, “thou fair death sword”, and “thou fair noble fair death sword” are in L . However, “fair noble sword” (does not begin with thou), “thou noble noble sword” (noble appears more than once), and “thou death sword” (death is not immediately preceded by fair) are not in L .

Part a Regular expression

Write a regular expression that recognizes L . You may only use concatenation, alternation ($|$), repetition ($*$, meaning zero or more repetitions), and emptiness (ε , the empty string).

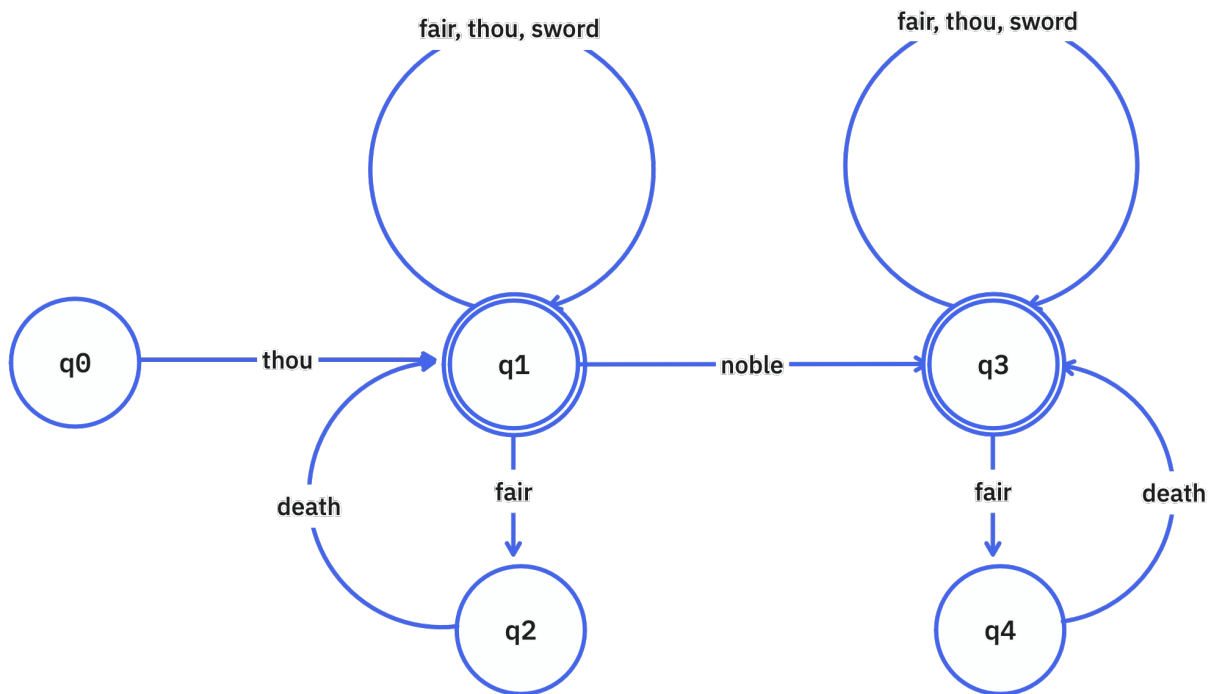
Answer

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thou (fair death | fair | thou | sword)*  
    ( $\varepsilon$  | noble)  
    (fair death | fair | thou | sword)*
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Problem 1 Part b

Give an NFA that recognizes the language L from Part a. You may assume that there is an implicit failure state and any omitted transitions automatically go to that failure state.

Answer



Start state: q_0 . Accepting states: q_1 and q_3 . Omitted transitions fail.

Problem 1 Part c

Consider the grammar for a language of logical formulas called P. This language consists of formulas composed of variables, conjunction ($\wedge$), disjunction ($\vee$), negation ($\neg$), and universal quantification ($\forall$). More precisely, this language is recognized by the following context-free grammar:

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<formula>    ::= <quantifier> <formula> | <sentence>
<quantifier> ::=  $\forall$  <var> .
<sentence>   ::= <clause> | <sentence>  $\wedge$  <clause>
<clause>    ::= <literal> | <literal>  $\vee$  <clause>
               | ( <clause> )
<literal>   ::= <var> |  $\neg$  <var>
<var>       ::= A | B | C | ... | Z

```

For each of the following examples, determine whether or not the given formula belongs in P (yes or no). If no, give a brief explanation of at most one sentence.

Formula	In P?	Explanation
$\forall X. \forall Y. (A \vee B) \wedge (\neg C \vee D)$	Yes	
$\forall X. (P \wedge Q) \vee (R \wedge S)$	No	Conjunctions may not appear within clauses.
$(A \vee B) \wedge \forall Y. (C \vee D)$	No	All quantifiers must appear before the sentence.
$D \vee \neg E \wedge F \vee G$	Yes	
Z	Yes	
$\neg(\neg W)$	No	Negation applies directly to a variable, not a parenthesized expression.
ε	No	The grammar requires at least one literal.